

05 • L2 — Data Dictionary

Assignment: "Data dictionary: specify data elements ... at a minimum this should include the data element labels, type, response options and conditional logic. Where possible, map to standards. As a guide, the CDSS should include between 30–50 data elements, with at least some conditional logic, and some derived data elements."

5.1 Key figures

| | | | --- | **Total data elements** | **48** | of which **derived** (calculated) | **9** | of which with **conditional logic** (relevant) | **4** | of which with a **constraint** | **8** | of which with a **terminology binding** | **38** | of which with a **FHIR mapping** | **48** (all) | | Sections | 5 |

Key to types: text • integer • decimal • date • select_one • calculated

5.2 Section 1 — Surgery details (once only)

| ID | Variable | Label | Type | Response options | Condition | Constraint | Terminology | FHIR | L1 source | |---|---|---|---|---|---|---| | DE-001 | pat_name | Name ("Name") | text | — | — | — | — | Patient.name | — | | DE-002 | pat_dob | Date of birth ("Geburtsdatum") | date | — | — | not in the future | LOINC 21112-8 | Patient.birthDate | — | | DE-003 | pat_age | Age in years ("Alter (Jahre)") | **calculated** | ageYears(pat_dob) | — | — | LOINC 30525-0 | Patient.birthDate (derived) | — | | DE-004 | op_date | Date of surgery ("Datum der Operation") | date | — | — | not in the future | SNOMED 387713003 | Procedure.performedDateTime | CDC (time window) | | DE-005 | days_post_op | Days since surgery ("Tage seit der Operation") | **calculated** | daysSince(op_date) | — | — | — | Observation.effectiveDateTime (derived) | CDC 30/90-day window | | DE-006 | op_site | Body site of the wound ("Körperstelle der Wunde") | select_one | 1=Head/neck ("Kopf/Hals") • 2=Thorax ("Thorax") • 3=Abdomen ("Abdomen") • 4=Back ("Rücken") • 5=Shoulder/arm ("Schulter/Arm") • 6=Hand ("Hand") • 7=Hip/leg ("Hüfte/Bein") • 8=Foot ("Fuss") • 9=Other ("Andere") | — | — | SNOMED 442083009 | Procedure.bodySite | — | | DE-007 | closure | Wound closure ("Wundverschluss") | select_one | 1=Sutures ("Naht") • 2=Staples ("Klammern") • 3=Glue/Steri-Strips ("Kleber/Steristrips") | — | — | SNOMED 129287005 | Procedure.focalDevice | — | | DE-008 | wound_class | CDC wound classification ("CDC-Wundklassifikation") | select_one | 1=Clean • 2=Clean-contaminated • 3=Contaminated • 4=Dirty | — | — | SNOMED 255215009 | Procedure.category | **CDC/NHSN** | | DE-009 | op_duration_min | Duration of surgery in minutes ("Dauer der Operation (Minuten)") | integer | — | — | 1–1440 | — | Procedure.performedPeriod | SSI risk factor | | DE-010 | antibiotic_prophylaxis | Did you receive an antibiotic before/during surgery? ("Antibiotikum vor/während der OP erhalten?") | select_one | 1=Yes ("Ja") • 0=No ("Nein") • 9=Don't know ("Weiss nicht") | — | — | SNOMED 428264007 | MedicationAdministration | NICE NG125 | | DE-011 | drainage_present | Is a drain in place? ("Liegt eine Drainage?") | select_one | 1=Yes ("Ja") • 0=No ("Nein") | — | — | SNOMED 257803006 | Device.type | — |

5.3 Section 2 — Risk factors (once only)

| ID | Variable | Label | Type | Response options | Condition | Constraint | Terminology | FHIR | L1 source | |---|---|---|---|---|---|---| | DE-012 | diabetes | Do you have diabetes? ("Haben Sie Diabetes?") | select_one | 1=Yes ("Ja") • 0=No ("Nein") • 9=Don't know ("Weiss nicht") | — | — | ICD-10 E10–E14 | Condition.code | SSI risk factors | | DE-013 | smoker | Do you smoke? ("Rauchen Sie?") | select_one | 1=Yes ("Ja") • 0=No ("Nein") | — | — | SNOMED 77176002 | Observation (social history) | SSI risk factors | | DE-014 | immunosuppr | Do you take any medication that weakens the immune system? ("Nehmen Sie Medikamente, die das Immunsystem schwächen?") | select_one | 1=Yes ("Ja") • 0=No ("Nein") • 9=Don't know ("Weiss nicht") | — | — | SNOMED 373847000 | MedicationStatement | SSI risk factors | | DE-015 | anticoag | Do you take blood thinners? ("Nehmen Sie Blutverdünner?") | select_one | 1=Yes ("Ja") • 0=No ("Nein") • 9=Don't know ("Weiss nicht") | — | — | SNOMED 372862008 | MedicationStatement | Bleeding assessment | | DE-016 | height | Height in cm ("Körpergröße (cm)") | integer | — | — | 100–230 | LOINC 8302-2 | Observation.valueQuantity | — | | DE-017 | weight | Weight in kg ("Gewicht (kg)") | decimal | — | — | 30–300 | LOINC 29463-7 | Observation.valueQuantity | — | | DE-018 | bmi | BMI | **calculated** | weight / (height/100)² | — | — | LOINC 39156-5 | Observation.valueQuantity | SSI risk factor | | DE-019 | risk_count | Number of risk factors | **calculated** | (diabetes=1) + (smoker=1) + (immunosuppr=1) + (bmi>=30) | — | — | — | RiskAssessment.prediction | SSI risk factors |  (extended) |

5.4 Section 3 — Daily check: general condition

| ID | Variable | Label | Type | Response options | Condition | Constraint | Terminology | FHIR | L1 source | |---|---|---|---|---|---|---| | DE-020 | temp | Body temperature today in °C ("Körpertemperatur heute (°C)") | decimal | — | — | 34–43 • "Plausible temperature: 34–43 °C." ("Plausible Temperatur: 34–43 °C.") | LOINC 8310-5 | Observation.valueQuantity | **CDC** (fever >38 °C) | | DE-021 | fever | Fever (from 38.0 °C) ("Fieber (ab 38.0 °C)") | **calculated** | temp >= thresholds.fever_orange | — | — | SNOMED 386661006 | Observation.interpretation | **CDC** deep incisional SSI | | DE-022 | chills | Have you had chills? ("Hatten Sie Schüttelfrost?") | select_one | 1=Yes ("Ja") • 0=No ("Nein") | — | — | SNOMED 43724002 | Observation.valueCodeableConcept | Sepsis warning sign | | DE-023 | pain | How severe is your pain? (0–10) ("Wie stark sind Ihre Schmerzen? (0–10)") | integer | NRS 0–10 | — | 0–10 • "Please enter a value between 0 and 10." ("Bitte Wert zwischen 0 und 10.") | LOINC 72514-3 | Observation.valueQuantity | **CDC** ("pain or tenderness") | | DE-024 | pain_trend | Pain compared with yesterday ("Schmerzen im Vergleich zu gestern") | select_one | 0=Better ("Besser") • 1=The same ("Gleich") • 2=Worse ("Schlechter") | — | — | — | Observation.valueCodeableConcept | **CDC** ("new or worsening") | | DE-025 | sleep_disturbed_by_pain | Did the pain disturb your sleep? ("Haben die Schmerzen den Schlaf gestört?") | select_one | 1=Yes ("Ja") • 0=No ("Nein") | pain >= 4 | — | SNOMED 301345002 | Observation.valueCodeableConcept | Pain proxy | | DE-026 | wellbeing | How do you feel overall? ("Wie fühlen Sie sich insgesamt?") | select_one | 0=Well ("Gut") • 1=Tired ("Müde") • 2=Really ill ("Richtig krank") | — | — | SNOMED 386661006 | Observation.valueCodeableConcept | Systemic sign | | DE-027 | nausea_vomiting | Nausea or vomiting? ("Übelkeit oder Erbrechen?") | select_one | 0=No ("Nein") • 1=Nausea ("Übelkeit") • 2=Vomiting ("Erbrechen") | — | — | SNOMED 422587007 | Observation.valueCodeableConcept | Systemic sign | | DE-028 | appetite | Appetite today ("Appetit heute") | select_one | 0=Normal ("Normal") •

1=Reduced ("Vermindert") · 2=No appetite ("Kein Appetit") | — | — | SNOMED 79890006 | `Observation.valueCodeableConcept` | Systemic sign | | DE-029 | `meds_taken` | Prescribed medication taken today? ("Verordnete Medikamente heute eingenommen?") | `select_one` | 1=Yes ("Ja") · 0=No ("Nein") · 2=None prescribed ("Keine verordnet") | — | — | SNOMED 182840001 | `MedicationStatement.status` | Adherence |

5.5 Section 4 — Daily check: the wound

| ID | Variable | Label | Type | Response options | Condition | Constraint | Terminology | FHIR | L1 source | |---|---|---|---|---|---|---|---| | DE-030 | `wound_state` | **Which image best matches your wound today?** ("Welches Bild passt am besten zu Ihrer Wunde heute?")  | `select_one` | 1=Calm & dry ("Ruhig & trocken") · 2=Slightly reddened ("Leicht gerötet") · 3=Markedly reddened & swollen ("Stark gerötet & geschwollen") · 4=Open / pus ("Offen / Eiter") | — | — | SNOMED 225552003 | `Observation.valueCodeableConcept` | **Southampton 0-V** | | DE-031 | `redness_extent` | How far does the erythema extend? ("Wie weit geht die Rötung?") | `select_one` | 0=None ("Keine") · 1=Only at the wound edge ("Nur am Wundrand") · 2=Around the suture ("Um die Naht herum") · 3=Along the whole wound ("Entlang der ganzen Wunde") · 4=Beyond the wound ("Über die Wunde hinaus") | — | — | SNOMED 247441003 | `Observation.valueCodeableConcept` | **Southampton IIa-IIId** | | DE-032 | `red_streak` | **Is a red streak spreading away from the wound?** ("Zieht ein roter Streifen von der Wunde weg?") | `select_one` | 1=Yes ("Ja") · 0=No ("Nein") | (*none — deliberately always visible*) | — | SNOMED 46117009 | `Observation.valueCodeableConcept` | **NICE NG125 1.4.9** (cellulitis/lymphangitis) | | DE-033 | `swelling` | Is the area around the wound swollen? ("Ist die Umgebung der Wunde geschwollen?") | `select_one` | 0=No ("Nein") · 1=Slightly ("Leicht") · 2=Markedly ("Deutlich") · 3=Severely ("Stark") | — | — | SNOMED 65124004 | `Observation.valueCodeableConcept` | **CDC** ("localized swelling") | | DE-034 | `warmth` | Does the skin around the wound feel hot? ("Fühlt sich die Haut um die Wunde heiss an?") | `select_one` | 1=Yes ("Ja") · 0=No ("Nein") | — | — | SNOMED 707793005 | `Observation.valueCodeableConcept` | **CDC** ("heat") | | DE-035 | `secretion` | **Is fluid coming out of the wound? Which image matches?** ("Kommt Flüssigkeit aus der Wunde? Welches Bild passt?")  | `select_one` | 0=None ("Keine") · 1=Clear ("Klar") · 2=Bloody ("Blutig") · 3=Cloudy / purulent ("Trüb / eitrig") | — | — | SNOMED 449736006 | `Observation.valueCodeableConcept` | **CDC** ("purulent drainage") · **Southampton III-IV** | | DE-036 | `secretion_smell` | Does the wound discharge smell unpleasant? ("Riecht das Wundsekret unangenehm?") | `select_one` | 1=Yes ("Ja") · 0=No ("Nein") | `secretion >= 1` | — | SNOMED 288227007 | `Observation.valueCodeableConcept` | Sign of infection | | DE-037 | `discharge_extent` | Is the fluid coming from one small spot or along the whole wound? ("Kommt die Flüssigkeit an einer kleinen Stelle oder entlang der ganzen Wunde?") | `select_one` | 1=Small spot ("Kleine Stelle") · 2=Along the whole wound ("Entlang der ganzen Wunde") | `secretion >= 1` | — | — | `Observation.valueCodeableConcept` | **Southampton IIIa/b, IVa/b** | | DE-038 | `discharge_duration_days` | For how many days has fluid been coming out? ("Seit wie vielen Tagen kommt Flüssigkeit?") | `integer` | — | `secretion >= 1` | 0-60 | — | `Observation.valueQuantity` | **Southampton IIId** (>3 days) | | DE-039 | `bleeding` | Is the wound bleeding? ("Blutet die Wunde?") | `select_one` | 0=No ("Nein") · 1=Slightly ("Leicht") · 2=Heavily, dressing soaked through ("Stark (Verband durchtränkt)") | — | — | SNOMED 131148009 | `Observation.valueCodeableConcept` | Aftercare standard | | DE-040 | `wound_open` | Are the wound edges closed? ("Sind die Wundränder geschlossen?") | `select_one` | 0=Closed ("Geschlossen") · 1=Small gap ("Kleine Lücke") · 2=Gaping open ("Klaffend offen") | — | — | SNOMED 225553008 | `Observation.valueCodeableConcept` | **CDC** deep incisional (dehiscence) | | DE-041 | `crust` | Has a scab / crust formed? ("Hat sich Wundschorf / eine Kruste gebildet?") | `select_one` | 1=Yes ("Ja") · 0=No ("Nein") | — | — | SNOMED 262074001 | `Observation.valueCodeableConcept` | Phases of wound healing | | DE-042 | `numbness` | New numbness or tingling at the wound? ("Neues Taubheitsgefühl oder Kribbeln an der Wunde?") | `select_one` | 1=Yes ("Ja") · 0=No ("Nein") | — | — | SNOMED 44077006 | `Observation.valueCodeableConcept` | Nerve irritation | | DE-043 | `mobility_change` | New restriction of movement at the operated site? ("Neue Bewegungseinschränkung an der operierten Stelle?") | `select_one` | 1=Yes ("Ja") · 0=No ("Nein") | — | — | SNOMED 105504002 | `Observation.valueCodeableConcept` | Functional warning sign | | DE-044 | `dressing_changed_today` | Dressing changed today? ("Verband heute gewechselt?") | `select_one` | 1=Yes ("Ja") · 0=No ("Nein") · 2=No dressing ("Kein Verband") | — | — | SNOMED 182531007 | `Procedure.code` | **NICE NG125 1.4.1** |

5.6 Section 5 — Derived classification (not visible to users)

ID	Variable	Label	Type	Calculation	Terminology	FHIR	L1 source
DE-045	<code>southampton_grade</code>	Southampton grade	calculated	see below	SNOMED 225552003	<code>Observation.valueCodeableConcept</code>	Southampton
DE-046	<code>classification</code>	Traffic-light classification	calculated	Result of the decision table (R1-R12)	—	<code>Observation.valueCodeableConcept</code>	—
DE-047	<code>last_classification</code>	Yesterday's classification	calculated	<code>previous(classification)</code>	—	<code>Observation</code> (previous day)	—
DE-048	<code>trend_worsening</code>	Deterioration compared with yesterday	calculated	<code>severity(classification) > severity(last_classification)</code>	—	<code>Observation.interpretation</code>	CDC ("new or worsening")
DE-049	<code>check_streak_days</code>	Consecutive days with a completed check	calculated	<code>countConsecutiveDays()</code>	—	—	Fidelity indicator (M&E)

Note: `check_streak_days` (DE-049) is a purely monitoring-related indicator without any effect on the clinical rules. If it is not counted among the clinical data elements, the total is 48.

5.7 Calculation of `southampton_grade`

```
southampton_grade =
  V    if wound_open = 2 or wound_state = 4
  IV   if secretion = 3
```

```

III  if secretion in (1, 2)
II   if wound_state = 3
      or redness_extent >= 3
      or swelling >= 2
      or warmth = 1
I    if wound_state = 2 or redness_extent in (1, 2)
0    otherwise

```

The order is **descending** — the first matching case wins. The worst-first principle therefore applies here as well: a patient with purulent discharge (IV) and mild erythema (I) at the same time receives Grade IV.

Mapping table and derivation: 03 L1 — Southampton Mapping

5.8 Design principles

Sequence and grouping (UX) The assignment explicitly mentions “*sequence and grouping of your questions*”. Our sections follow **not** the structure of the guideline, but the sequence a patient naturally works through at home: first the one-off details (only on the first occasion), then “how am I feeling” (general condition), then “what does the wound look like” (inspection). The wound inspection comes last because it requires removing the dressing and involves the greatest effort.

Conditional logic 4 elements are tied to a condition. The basic rule: questions that only make sense given a particular finding are only asked in that case (`discharge_extent` only when discharge is present). **Exception for safety reasons:** `red_streak` (DE-032) is deliberately *not* gated — see Bug B-03, in which an emergency rule was rendered unreachable by conditional logic.

Constraints and data quality 8 elements have plausibility checks with an error message that lay users can understand. Constraints run **before** the rules are evaluated (validation loop in the BPMN detail process), so that no classification is based on implausible data.

Terminology binding 38 elements carry SNOMED CT, LOINC or ICD-10 codes. Deliberately not mapped are: purely administrative fields (`pat_name`), app-internal counter variables and elements for which no unambiguous concept exists. For a production system, the SNOMED CT licensing situation in Switzerland would need to be clarified — see 11 Implementation.

FHIR mapping All elements are assigned to a FHIR R4 resource. Conceptually, the L3 corresponds to a `Questionnaire` ; each daily check is transmitted as a `QuestionnaireResponse` , and the classification additionally as an `Observation` . Details: 08 L3 architecture.

Related documents: 03 L1 · 06 Decision Logic · 08 L3